Attention-Based Deep Neural Architectures for Sequence Transduction and Semantic Representation

Ashish Vaswani, Elena Chen, Rajesh Mercer

Department of Computer Science, Institute for Advanced Artificial Intelligence

ABSTRACT

1. INTRODUCTION

Recurrent neural networks, long short-term memory (LSTM), and gated recurrent neural networks have been firmly established as state-of-the-art approaches in sequence modeling. However, sequential computation remains an inherent bottleneck in recurrent language models, preventing parallelization within training examples. This fundamental limitation becomes critical at longer sequence lengths, as memory constraints limit batching across examples.

Recent work has achieved remarkable improvements in computational efficiency through factorization tricks and conditional computation, but the fundamental constraint of sequential execution remains. Attention mechanisms have become an integral part of compelling sequence modeling, allowing modeling of dependencies without regard to their distance in input sequences.

In this work, we propose the Transformer, a model architecture eschewing recurrence and instead relying entirely on an attention mechanism to draw global dependencies between input and output representations. The Transformer allows for significantly more parallelization and can reach a new state of the art in translation quality after being trained for as little as twelve hours.

2. MODEL ARCHITECTURE AND METHODOLOGY

Most competitive neural sequence transduction models have an encoder-decoder structure. Here, the encoder maps an input sequence of symbol representations to a sequence of continuous representations. Given this continuous representation, the decoder then generates an output sequence of symbols one element at a time.

Our proposed architecture follows this overall framework using stacked self-attention and point-wise, fully connected layers for both the encoder and decoder. The encoder is composed of a stack of N identical layers. Each layer has two sub-layers: a multi-head self-attention mechanism, and a simple, position-wise fully connected feed-forward network.

An attention function can be described as mapping a query and a set of key-value pairs to an output. The output is computed as a weighted sum of the values, where the weight assigned to each value is computed by a compatibility function of the query with the corresponding key. Specifically, we compute scaled dot-product attention over query matrix Q, key matrix K, and value matrix V.

Multi-head attention allows the model to jointly attend to information from different representation subspaces at different positions. With a single attention head, averaging inhibits this capacity. In addition, we apply residual connections around each of the sub-layers, followed by layer normalization to ensure gradient stability throughout training.

3. EXPERIMENTAL EVALUATION AND RESULTS

On the WMT 2014 English-to-German translation task, the big transformer model out-performs the best previously reported models (including ensembles) by more than 2.0 BLEU, establishing a new state-of-the-art BLEU score of 28.4. On the WMT 2014 English-to-French translation task, our big model achieves a BLEU score of 41.0, outperforming all previously published single models at 1/4 the training cost.

Training was carried out on standard NVIDIA P100 GPUs using the Adam optimizer with beta1 = 0.9, beta2 = 0.98, and epsilon = 10^-9. We varied the learning rate over the course of training according to a warm-up schedule. Model checkpoints were averaged over the last five evaluations to produce the final inference parameters.

4. CONCLUSION AND FUTURE WORK

In this work, we presented the Transformer, the first sequence transduction model based entirely on attention, replacing recurrent layers most commonly used in encoder-decoder architectures with multi-head self-attention. For translation tasks, the Transformer can be trained significantly faster than architectures based on recurrent or convolutional layers.

The primary contribution of this research is establishing that attention alone is sufficient for sequence modeling, achieving superior translation accuracy while eliminating sequential execution bottlenecks. We plan to extend the Transformer to problems involving input and output modalities other than text and investigate local, restricted attention mechanisms to efficiently handle massive inputs such as images and audio.

5. REFERENCES

[1] Bahdanau, D., Cho, K., & Bengio, Y. (2015). Neural machine translation by jointly learning to align and translate. ICLR.
[2] Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. Neural Computation, 9(8), 1735-1780.
[3] Sutskever, I., Vinyals, O., & Le, Q. V. (2014). Sequence to sequence learning with neural networks. NeurIPS.
[4] Kingma, D. P., & Ba, J. (2015). Adam: A method for stochastic optimization. ICLR.